

JANUARY 2024

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

i. Which point on the stress strain curve occurs after the ultimate point?

- a) Last point b) Breaking point
 c) Elastic limit d) Material limit

ii. The hinged end is also known as _____

- a) Fixed end b) Pinned end
 c) Rigid end d) Free end

iii. The slenderness ratio is the ratio of effective length to least _____

- a) Ultimate load b) Actual length
 c) Radius of gyration d) Factor of safety

iv. _____ is half the circumferential stress.

- a) Hoop stress b) Longitudinal stress
 c) Fluid stress d) Transverse stress

v. What are the units of Polar modulus?

- a) mm^3 b) mm^2
 c) mm d) mm^4

vi. U.D.L stands for?

- a) Uniformly diluted length b) Uniformly developed loads
 c) Uniaxial distributed load d) Uniformly distributed loads

vii. Units of U.D.L?

- a) KN/m b) KN-m
 c) $\text{KN-m} \times \text{m}$ d) KN

viii. At the Point of contraflexure, what is the value of bending moment?

- a) one b) zero
 c) three d) infinity

ix. When SF is zero, the bending moment is _____

- a) Zero b) Maximum
 c) Very difficult to say d) Minimum

x. What is the moment of inertia of a rectangular section about an horizontal axis through C.G?

- a) $bd^3/6$ b) $bd^2/12$
c) $b^2d^2/12$ d) $bd^3/12$

xi. What is the section modulus (Z) for a rectangular section?

- a) $bd^2/6$ b) $a^3/6$
c) BD^3-bd^3 d) D^4-d^4

xii. On bending of a beam, which is the layer which is neither elongated nor shortened?

- a) Axis of load b) Neutral axis
c) Center of gravity d) None of the mentioned

xiii. In cantilever beam the deflection occurs at _____

- a) Free end b) Point of loading
c) Through out d) Fixed end

xiv. The angle of twist can be written as _____

- a) TL/J b) CJ/TL
c) TL/CJ d) T/J

xv. Shear force is unbalanced _____ to the left or right of the section.

- a) Horizontal force b) Vertical force c) Inclined force d) Conditional force

Group-B

2. Draw the stress-strain diagram for a mild steel specimen. Show the salient points in the diagram (8)

3. A cantilever beam AB, 2m long carries a uniformly distributed load of 1.5 kN/m over a length of 1.6 m from the free end. Draw shear force and bending moment diagrams for the beam. (8)

4. Find the moment of inertia of a T-section with flange as 150 mm x 50 mm and web as 150 mm x 50 mm about X-X and Y-Y axes through the centre of gravity of the section. (8)

5. Calculate the maximum torque that a shaft of 225 mm diameter can transmit, if the maximum angle of twist is 1° in a length of 1.5 m. Take $C = 70 \times 10^3 \text{ N/mm}^2$ (8)

6. A steel rod 5 m long and of 40 mm diameter is used as a column, with both end hinged. Determine the crippling load by Euler's formula. Take $E = 200 \times 10^3 \text{ N/mm}^2$ (8)

7. A closely-coiled helical spring of round steel wire 5 mm in diameter having 12 complete coils of 50 mm mean diameter is subjected to an axial load of 100 N. Find the deflection of the spring and the maximum shearing stress in the material. Modulus of rigidity (C) = $80 \times 10^3 \text{ N/mm}^2$ (8)

8. A steam boiler of 800mm diameter is made up of 10 mm thick plates. If the boiler is subjected to an internal pressure of 2.5 MPa. Find the circumferential and longitudinal stresses induced in the boiler plates. (8)

9. Write short notes on the followings (Any four) a) Elasticity b) Malleability c) Fatigue d) Creep e) Poisson's Ratio f) Brittleness (4×2 = 8)

10. Prove that $T/J = C.\theta/L$ for a solid circular shaft. What are the assumptions made in derivation of torsion equation. (5 + 3 = 8)